COMP 361

Course Project

Multiple Due Dates

Introduction

COMP 361 is populated by Computer Science majors (only), Computer Science majors with a Software Engineering Specialty, and Computer Science majors with a Games Specialty (if you are another type of major, you are treated as a Computer Science major only). Each group has different course requirements. To support these differences, the course project is divided into two parts: (a) the Base Project and (b) the Special Themes.

Group 1:
Computer Science majors (only) and Computer Science majors with a Software Engineering Specialty are required to complete either the Base Project (only) or the Base Project + Specialty Theme #1.

Group 2:
Computer Science majors with a Games Specialty are required to complete the Base Project + Specialty Theme #2 or the Base Project + Specialty Theme #3. The project must be built using a game engine: Unity or Unreal.

Group 3:
Students in Group #1 who end up on a team with Group #2 students can switch allegiance and work under the Group #2 rules. The opposite is not true: Game students MUST do a game in an Engine.

Team Formation

Your team must have 7 students. If there are not enough students to evenly divide the class into groups of 7, one group may have 6 or 8 students. Speak to the professor first to get approval for a team size other than 7. Assignment #1 directs you to the form you will need to fill out to report to the professor your team members, the captain, and the project theme you have selected.

How to find a team:

- Start with people you know in class.
- Talk to random people near you: “What theme is your project?” “I’m looking for a team”
- Post on uAsk with #project.
- Come to class because the prof will randomly announce, “Who needs a team?” and begin matchmaking.
- Contact the McGill Games Club: https://gamedevmcgill.ca/

Project Themes

Based on the group you belong to, you will need to decide on the type of project you and your team will construct. This is a two-term project. The first term will focus on designing the software and learning to program using the tools and engineering techniques specific to your project. You will need to self-study the libraries, languages, and engines you will be using. In the second term, you will start

programming (officially).

The project is built around a base architecture that you can modify using a Theme. The base architecture is called the Base Project, and there are three theme modifiers. In the introduction to this document, I described the student population and the group they belong to, and the rules for switching allegiance. Please review that before reading below:

Base Project:

- Base Application: A desktop application program in any language you like.
- Base Theme:
  - Propose a project based on the theme "App to help students, a club, or McGill."
  - Get the approval of the professor before registering your group (deductions for not following instructions).
- Minimum Features:
  - Graphical user interface (not only a text-based interface).
  - Database or some type of input/output file system (persistent storage).
  - Security feature(s).
  - Cannot be a website (that is COMP 307).
  - Each team member:
    - As a minimum, each team member is responsible for at least one module of the application program, where a module is understood to be: (i) one or more GUI interfaces (like a window or screen – if one interface is a GUI then the second interface can be something else, like text-based or an API), (ii) reading/writing to the database or file system, (iii) more than one object handling the business logic for the interface, (iv) following design pattern rules, (v) must provide automated test cases for their code, and (vi) the code should be well documented (comments and a readme file).
  - As a team:
    - Identify the Patterns and Anti-patterns the application program is based upon.
    - Select a common development stack: IDE, Repository, Code documentation standards, Automated testing method, and software management tools.
    - Select a deployment environment that is the "official" runtime platform for the project (maybe this would be the team captain's laptop, or it could be the Mimi server, or something else). That means if the module you are responsible for works well on your laptop but not on the official runtime platform, you won't be graded for it. A module is only considered working when it runs on the official runtime platform.
    - During demos, the program can only be demonstrated from a single laptop: the official runtime platform.
  - You can add additional elements to the application program after receiving approval from the professor.

Theme #1: Hex Game Project

- The Hex Game Project is the Base Project but swapping out the Base Theme for Theme #1.
- The Theme #1 theme: Create a video game that uses a hex-based map for a game that is loosely based on Dungeons & Dragons.
- Restrictions: You cannot use a game engine, but you can use a graphics library. For example, in Java you could use Swing.

Theme #2: Game Engine Project

- The Hex Game Project is the Base Project but swapping out the Base Theme for Theme #2.
- The Theme #2 theme: Create a video game using either the Unreal or Unity game engines for a game that is loosely based on Dungeons & Dragons.

Theme #3: Ubisoft Competition

- The Ubisoft competition begins and ends only in our second semester. https://montreal.ubisoft.com/en/our-commitments/education/game-lab-competition/ We will not know what Ubisoft will want you to build until late January, maybe even early February. Given this restriction and the requirements of COMP 361, your project will have to look like this:
    - Fall Semester: You must do Theme #2, but:
        - You will design a D&D game like everyone else in the class, completing the assignments as if you are not doing the competition.
        - In parallel:
            - You will learn to program Unity or Unreal to prepare yourself for the competition.
            - You will begin to create assets and scripts to prepare for the competition theme reveal, which is always a surprise. The professor will help guide you with this part. Make an appointment with the professor.
    - Winter Semester:
        - You must meet with the professor to plan out your Winter term.
        - You will receive Ubisoft Mentors at some point and will need to attend various Ubisoft-required meetings and events.
        - You will have to submit the Scrum Journal, but no other reports.

Submissions

This project is divided into a series of submissions. Each submission will help you progress toward your end goal: building a fully functional application while managing and working in a team of seven developers.

THE FALL SEMESTER

Assignment #1 – Team Formation and Project Selection
Details will be defined within the assignment description. At this stage, your job is to find your team members and choose the project theme you want to work on. You will also need to confirm your choices with the professor and register your team.

Assignment #2 – Artifacts for the Specification Document
Details will be defined within the assignment description. The first stage of any project is to determine the application's mechanics and how the user will interact with it. From this, you will begin to glimpse the application's technology stack. You will be asked to create the following diagrams: a UML Use case diagram and Template, the application Storyboard, Wireframes, the Initial technology stack, and product owner win and loss conditions (the professor is not the product owner), among other things (business rules, effort estimation).

Assignment #3 – Artifacts for the Design Document

Details will be defined within the assignment description. After the product owner's OKAY and the product Freeze, we can move into designing the software. Design patterns, final technology stack, libraries, programming language(s), etc. UML Domain Models, Activity Diagrams, Sequence Diagrams, Deployment Diagrams, Database schemas, some screen mock-ups, critical sections, etc.

Assignment #4 – Artifacts for the Development Document
Details will be defined within the assignment description. IDE, Test cases, Quality Assurance, Repository pipelines, Agile, Scrum, customer-in-the-loop, management tools, templates, scheduling, deployment environment, and teamwork assignment (responsibilities), etc.

THE WINTER SEMESTER

Three Progress Meetings
Every second week of the month, your team will meet with either the TA or the Professor (your "handler") to discuss your progress. This is graded. Each team member is graded individually based on their contribution to project **planning/research** at this meeting.

Three Progress Demos
Every fourth week of the month, your team meets with your handler to see a demo of the program in its current state. This is graded. Each team member is graded individually based on their contribution to the project's **code base** at this meeting.

Final Demonstration
This final demonstration happens when you are very close to completing your project. You will make an appointment with your handler to demo the completed application program. Your handler is not permitted to touch your program. One person from your team will demo the application to the handler. The handler will ask each team member questions to determine their level of contribution and knowledge of the program and the software engineering methods used. This is an individual grade.

Acceptance Test
Not everyone needs to attend this meeting. At this meeting, the team may not touch the program except to launch it. Then, they give the laptop to the handler. Team members may not speak, and the handler may not ask questions (unless there is an emergency). This is a group grade based on the quality of the application program. After this submission, the programming portion of the course has ended.

Scrum Meeting Journal
This is a log of all the meetings you had with your team **starting from assignment #2**. Each team member writes their own personal scrum journal. Submit the scrum journal on myCourses in the Assignment module. The journal is private. Only the professor will read it. The journal must have the following format:

Log each team meeting (min 2-4 per month)
- Summary of what was discussed (from your point of view (POV))
- Summary of decisions (from your POV)
- Your current reflection about the project and the team

End-of-Month Feelings: (1 per month)
- This is what I liked about this month
- This is what I am not liking about this month

- What are my next steps to fix personal and technical issues
- What are my next steps on the project

End-of-Term: (1 per term – two in total (Fall and Winter))
- What worked (personally, team-wise, technically)
- What did not work (personally, team-wise, technically)
- What I would do again in the future (personally, team-wise, technically)
- Personal reflections and feelings (conclusions and future)

Showcase

In the week before the Winter final exam season, we will hold a special class meeting to showcase everyone's project. **Attendance is graded**. We will set up tables with your deployment laptop running your application. One team member will be the host. Everyone else will walk around to look at and try out one another's applications. The host can be swapped out so they can tour the room, too. We will hold a secret ballot to vote for the best program. There will be a small prize (TBD).